

CityPulse AI

Code review, novelty audit and a founder's improvement plan

Prepared 1 October 2026 for Pragatish N, Ravi and Jyotish (VIT Chennai). Both code folders were read, never modified.

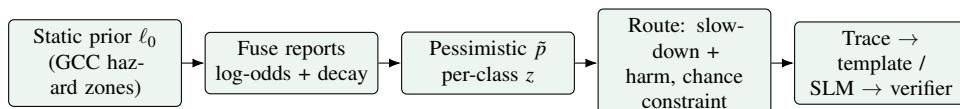
Bottom line. The research repository (`citypulse-IDP`) is a serious, well-engineered system with unusually honest documentation; the Kotlin app (`citypulse-ai`) is a demo that contradicts it in several places and should be frozen as a UI mock-up. Three of the four headline novelties do not survive the literature, and a fourth (the ALT “admissibility result”) is a known property from 2007. What *is* defensible is the combination: an on-device router whose caution grows as evidence thins, with explanations checked against the routing decision. An independent re-analysis of your own Study 1 data shows the current evaluation harness mis-measures routes, and that the default commuter setting changes only 7 of 100 routes. The fastest path to both a publishable paper and a real product is to (1) fix the evaluation, (2) change the cost model so short, dangerous segments are actually avoided, and (3) capture data from the northeast monsoon that is starting now.

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1 What CityPulse is, stated cleanly

CityPulse is hazard-aware navigation for Chennai that keeps working when the network does not. It keeps a probability for each road segment that it is affected by flooding (or another hazard class), built from a static susceptibility prior plus timestamped reports whose weight fades with age at a rate specific to the hazard type. It routes on a *pessimistic* version of that probability, so that roads with little evidence are treated more cautiously, with a caution dial z set per user class (commuter, emergency, pedestrian). It keeps travel slowdown (from a depth-disruption curve) separate from harm (a risk penalty), and removes edges outright when reported depth exceeds a threshold. Every route comes with a structured decision trace; an explanation is rendered from that trace by a template, optionally rewritten by a small on-device model or a cloud model, and shown only if a symbolic verifier finds every number, name and comparison grounded in the trace. All of this runs on the phone; the server only ingests reports and syncs.



2 What exists today

	citypulse-IDP (research repo)	citypulse-ai (Android, Kotlin)
Graph	Real Chennai OSM graph: 193,191 nodes, 471,240 edges; CSR + bidirectional Dijkstra + ALT	Hand-made 18-node, 26-edge toy graph
Belief	ℓ_0 from 7,453 GCC zones; log-odds fusion; per-class T_c ; $\tilde{p} = \min\{1, \bar{p} + z\sqrt{\bar{p}(1-\bar{p})/(n+1)}\}$	Similar fusion but $n_0 = 3$, clamps $\tilde{p} \in [0.01, 0.99]$, half-lives in minutes
Cost	$\tau_0[1 + \tilde{p}(\delta - 1)] + \lambda \tilde{p}s\tau_0$; δ from Pregmolato depth curve, clamped ≥ 1 ; chance constraint as edge removal	$\delta(\tilde{p}) = 1 + 3.2\tilde{p}^{1.5}$ attributed to Pregmolato (it is a depth function, not a probability function); severed = 100,000 s
Explanation	Template renderer + six-rule verifier (75 tests); Tier-1 flutter_gemma and Tier-2 Groq rewriters behind a 1.2 s deadline	“Tier-1 SLM” is a second template; verifier checks numerals only, with absolute values
Data	6,132-observation 2015 replay corpus; 402 watchlist candidates; deterministic replay harness	Six seeded hotspots
Offline/sync	SQLite + R*-tree cache, outbox, HLC, FastAPI server with GDACS/Open-Meteo ingest	Room DB with destructive migration
Evaluation	Study 1 (route quality) and Study 2 (calibration) run and honestly recorded (ADR-012)	None beyond unit tests

Recommendation: treat citypulse-IDP as the product and the paper. Keep the Kotlin app only for its visual design ideas; do not demo its numbers, because its confidence badge is hard-coded (94% online, 88% offline) and its verifier would pass “saves 3 minutes” when the trace says the route is 3 minutes *slower*.

3 Code review findings

Severity: **High** affects correctness of claims or safety; **Med** affects product quality or evaluation strength; **Low** hygiene.

#	Sev.	Where	Finding	Fix
1	High	study1_route_quality.py, query_orchestrator.dart	worst_edge_p is each configuration’s own $\tilde{p}$ under its own hazard set and z . C0 therefore scores 0, “exposure-reduction efficiency” comes out negative, and the Pareto y-axis rises with z by construction (36%→72%).	Score every route under one reference belief (done in Section 4); emit per-edge $\tilde{p}$ in the trace or re-score offline.
2	High	same	Detour ratio uses duration_s, which includes $\lambda \tilde{p}s\tau_0$ penalty seconds. C3’s reported mean detour of 0.27% is really 0.0013% in travel time.	Report free-flow (or expected) travel time separately from objective cost.
3	High	edge_cost.dart (model)	The penalty is proportional to τ_0 , so an edge can cost at most $(1 + \lambda s) \times$ free flow. With commuter $\lambda = 0.3$ the router is nearly hazard-blind (29% of routes still cross a $\tilde{p} \geq 0.5$ edge vs 30% for C0); a 50 m flooded underpass is almost free to traverse.	Add a traversal-independent harm term $\mu \tilde{p}s$ (seconds per edge) and/or a per-route risk budget; set λ, μ per class from stated preferences.
4	High	corpus + edge_cost.dart	depth_mm is null everywhere, so the chance constraint (the core safety mechanism) never fires on real data.	Collect categorical depth (ankle/knee/waist/impassable) in the report UI; fetch OpenCity “Inundation Points with Depth”.
5	Med	fusion.dart	Reports are summed as independent evidence; 5,052 crowd reports of one event are strongly correlated, so beliefs become over-confident.	Cap per-source/per-cluster weight, or estimate reliabilities with Dawid–Skene/truth discovery.
6	Med	pessimistic.dart	Wald-type band collapses as $\tilde{p} \rightarrow 0$ or 1; the “+1” mixes prior strength with evidence; n_{eff} ignores source reliability.	Use a Wilson/Beta-quantile bound with an explicit prior pseudo-count and reliability-weighted n_{eff} .
7	Med	cli_io.dart	One hazard class per edge; 124 edges lost their minority class.	Allow a list of class entries per edge; take the max cost or sum harms.
8	Med	CLI	3.07 s per call, almost all graph reload, which is why Study 1 used 100 instead of 1,000 pairs.	Add a batch mode or long-running process (JSON lines on stdin).
9	Med	app/assets/graph	47.5 MB JSON graph shipped as an asset (target was about 20 MB binary). Parse time and memory will hurt a Rs 10–15k phone.	Packed binary (FlatBuffers or raw typed arrays) loaded by memory map; tile the city.
10	Med	cloud_rewriter.dart	The cloud rewriter calls Groq from the client with a compile-time key (<code>-dart-define</code>), which anyone could extract from the APK. No key exists yet and the file itself flags this.	Proxy through the FastAPI server with rate limits (already noted as future work in the file).

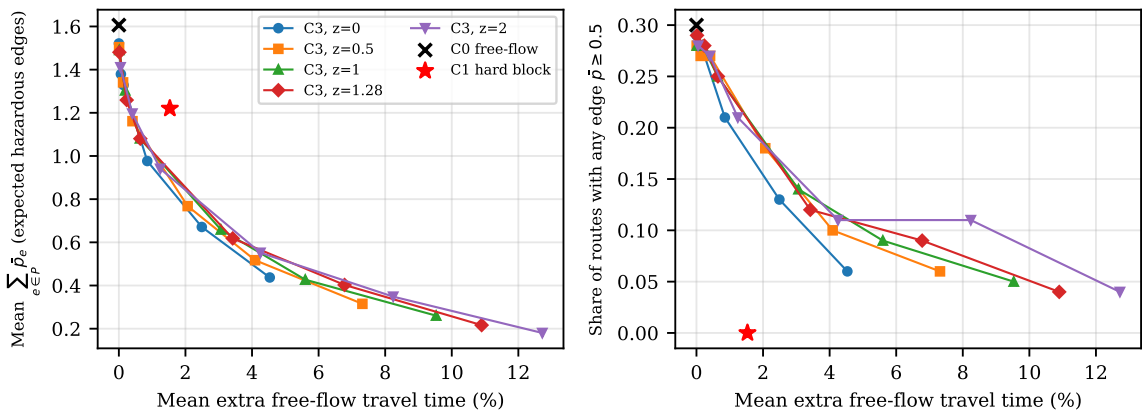
#	Sev.	Where	Finding	Fix
11	Med	verifier.dart	Numeral check is unit-agnostic (“4 km” is accepted if “4 minutes” is a trace fact); entity check depends on capitalised Latin words, so Tamil text cannot be verified.	Bind numbers to units and subjects via slot-filled templates; use a lexicon of trace entities instead of capitalisation.
12	Med	Study 2	The source-holdout label gives AUROC 0.609 and near-zero resolution; the Beta-reputation baseline is disadvantaged by lacking the prior.	Get independent labels (Section 8); give every baseline the same prior.
13	Med	prior (T1.3)	Category probabilities are planning defaults; no elevation or HAND; basin labels from a latitude-only heuristic; Mudichur corridor outside the GCC KML.	Add a DEM/HAND layer; fit category probabilities to 2015/2023 inundation; add Tambaram-area sources.
14	Low	Kotlin AppDatabase	<code>fallbackToDestructiveMigration()</code> wipes the offline cache on schema change; unknown categories silently become chronic waterlogging.	Explicit migrations; reject unknown categories.
15	Low	repo	Nested duplicate <code>citypulse-IDP/citypulse-IDP</code> ; the outer copy is stale (different <code>DECISIONS.md</code>).	Keep one root; archive the other.
16	Low	research/raw/A	The DOI given for Senarath et al. (Waze Bayesian fusion) resolves to an unrelated paper; “Liang et al. hazard-aware A*” is actually a paper on shortest paths and rescue-force dispatch.	Re-verify every citation before it enters a paper (this review did so for 129 works).

What is already excellent: one router implementation for app and evaluation; input validation that fails closed instead of letting a NaN keep a flooded edge open; determinism checks with SHA-256; ADRs that record negative results instead of hiding them; about 250 tests; strict legal hygiene (no Google Maps data, no OSM tile scraping, DPDP-aware retention).

4 Independent re-analysis of your Study 1 data

I re-scored the stored C0, C1 and C3 traces from `data/results/2026-09-18-study1-route-quality` under one common reference belief (C3’s fused $\bar{p}$ at the replay clock) and measured detours in free-flow time. A separate SciPy implementation of your cost function reproduced the Dart router’s C3 paths on 100/100 origin–destination pairs, and was then used to sweep $z \in \{0, 0.5, 1, 1.28, 2\}$ and $\lambda \in \{0.3, 1, 2, 5, 10, 20\}$.

Config.	z	λ	Mean extra time (%)	P90 extra time (%)	$\sum \bar{p}$ per route	Share $\bar{p} \geq 0.5$ (%)	TW-mean $\bar{p}$
C0 (free-flow)	–	–	0.00	0.00	1.61	30	0.0113
C1 (block $\bar{p} \geq 0.5$)	0	0	1.53	6.20	1.22	0	0.0078
C3	0	0.3	0.00	0.00	1.52	29	0.0100
C3	0	5	0.85	2.73	0.98	21	0.0062
C3	0	20	4.53	13.93	0.44	6	0.0023
C3	1	5	3.06	10.51	0.66	14	0.0037
C3	1	20	9.53	25.57	0.26	5	0.0009
C3	2	0.3	0.05	0.07	1.41	28	0.0094
C3	2	5	4.25	12.20	0.55	11	0.0028
C3	2	20	12.72	30.97	0.18	4	0.0005



Reading. (a) The commuter default changes almost nothing (7/100 routes, +0.026 s). (b) Larger λ buys exposure reductions smoothly; at small detours it beats hard blocking on expected exposure (0.98 at 0.85% vs 1.22 at 1.53%). (c) Hard blocking is the only setting that removes every high-probability edge. (d) Higher z does not help on this metric, and cannot: pessimism only pays off against ground truth the belief might be wrong about. This is why independent labels are the single most important missing asset.

5 Landscape and novelty audit

5.1 Who already does what

Actor	Relevance (sources in the literature review; product facts marked * come from your team’s research notes and were not re-verified here)
Academic flood routing	Dijkstra on flood models (Li et al. 2023), rescue dispatch (Liang et al. 2025), DRL rescue paths (Li & Wang 2025), dam-break evacuation (An et al. 2025). All assume a known flood state.
Patents	IBM flood-aware travel routing application (published 2013, abandoned*); Uber US10563994B2 on per-segment safety scores. Routing around hazards is prior art.
Crowd flood sensing	Waze flood reports assessed in Norfolk (Praharaj et al. 2021) and Dallas (Safaei-Moghadam et al. 2023); social-media flood monitoring (Wang et al. 2018).
Forecasting	GloFAS; Google’s global flood model (Nearing et al., <i>Nature</i> 2024) behind Flood Hub. River and basin scale, not streets.
Consumer and government apps	Google Maps and Waze (closures, report expiry)*, TomTom hazard feeds*, TN-ALERT (Tamil Nadu government, saved-location flood alerts in Tamil)*.
Dynamic routing theory	Landmark (ALT) search stays correct while no weight drops below its initial value (Delling & Wagner 2007).

5.2 Claim-by-claim verdict

Claim	Verdict	Defensible framing
Dynamic hazard-aware routing	Not novel	Cite academic work and patents; do not claim.
Confidence decay with report age	Not novel	Beta reputation with forgetting (2002), persistence filter (2016). Claim only <i>calibrated, class-specific</i> decay once you can fit it on time-stamped data.
Haven Mode	Not novel	TN-ALERT covers saved-place alerts*; keep rescope (ADR-006).
ALT admissibility under hazards	Known	Credit Delling & Wagner (2007); present as a property your cost guarantees via the $\delta \geq 1$ clamp.
Pessimism grows as evidence thins, per user class	Defensible	Framed as a design argument (confidence-multiplied penalties are risk-seeking under ambiguity) with a per-class dial.
On-device route + confidence + verified explanation	Defensible	“We are not aware of work combining...” once the verifier pass rate is measured on a low-cost phone.
Bounded deterrence of time-proportional penalties	New finding	Proposition + evidence from your own graph (this review).
Fair evaluation protocol for hazard routing	Useful	Common-reference scoring + event splits + released benchmark.

6 How to make CityPulse genuinely unique

- Traversal silence as evidence (the moat).** Every trip that passes a segment without a report is weak evidence it is passable. Model it as missing-not-at-random (drivers avoid roads they expect to be flooded). This is the one input dense exactly where users are, and nobody can buy it.
- Depth, not presence.** A four-level depth report (ankle, knee, waist, impassable) plus optional photo turns the dormant chance constraint on and lets pedestrian and two-wheeler thresholds differ from cars.
- Harm that does not scale with time.** A per-edge harm term so a 50 m flooded underpass is avoided even when the detour is long; this is a small change with a large, measurable effect (Section 4).
- Reservoir-to-corridor causality.** Chennai’s floods are partly driven by releases into the Adyar and Cooum basins. A basin graph that raises priors downstream of a reported release gives offline mode something to reason with that global products do not model at street level.
- “We do not know” as a first-class state.** Show data gaps differently from observed-clear roads; measure appropriate reliance, not satisfaction.
- Verifier as liability shield and training signal.** “Never assert passability” plus an audit log is what an insurer or a government partner will ask for; passing rewrites are free supervision for fine-tuning the small model.

7. **Tamil first.** Template-first explanation makes reliable Tamil output possible where small models are weak; it is also the equity argument.

7 The startup view

7.1 Problem and customers

- **Two-wheeler commuters and delivery riders** in Velachery, Pallikaranai, Mudichur and similar corridors, who today rely on WhatsApp groups and guesswork. High pain, zero willingness to pay, but they are the sensors.
- **Fleet operators** (quick commerce, food delivery, ride-hailing, intra-city logistics): lost orders and stranded vehicles every monsoon; they already pay for routing. First revenue.
- **Emergency services** (108 ambulances, fire): highest stakes, slow procurement; design partner rather than customer.
- **City and state agencies** (GCC, TNSDMA): buyers of a passability layer and dashboards via grants or MoUs; 12–24 month cycles.
- **Motor insurers:** flood-damage claims give a reason to pay for avoidance evidence (hypothesis to test).

7.2 Value proposition and model

Positioning: “Chennai’s monsoon passability layer, built from the vehicles already moving, with every routing claim checked before it is shown.” Free citizen app (the sensor network) → paid fleet API (per active vehicle per month, or per route call during alert days) → government and CSR licences for dashboards and data.

Why on-device matters commercially: your own cost model estimates that about 78% of a 1,000-user pilot’s cloud bill would be LLM calls; the on-device template path removes that cost and keeps working when towers fail.

7.3 Moat

Data network effect from traversal silence and depth reports; local basin knowledge; a verified-explanation stack that reduces liability; a public benchmark that makes you the reference others compare against.

7.4 Key metrics

- Coverage: share of watchlist segments with evidence younger than 1 h during an alert.
- Accuracy: Brier score and reliability on a held-out flood event; false-avoidance and missed-hazard rates against independent labels.
- Safety: share of routes crossing a segment later confirmed impassable.
- Explanation: verifier pass rate by tier; time to first explanation on a Rs 10–15k phone.
- Growth: reports per active user per alert day; fleet vehicles on the API; retention across two monsoons.

7.5 Risks

Risk	Mitigation
Cold start on flood day (no users, no reports)	Seed through one fleet and two resident associations before the first heavy rain; prior + reservoir signals as fallback; say “no data” plainly.
Liability if someone is harmed	Never assert passability (ADR-011); relative-risk wording; disclaimer gate; register an entity before any paid deployment.
Low-cost phones too slow for the SLM	Template is the product; SLM is optional; benchmark on the cheapest target device in week 1.
Data licences	OSM/ODbL only; no Google data; DPDP consent for any location use.
Retention outside monsoon	Integrate into daily commute or fleet dispatch; do not build a standalone “flood app”.

8 90-day roadmap (aligned with the northeast monsoon)

When	Deliverables
Week 0–1	Start data capture now: Open-Meteo, GDACS, CMWSSB lake levels (from an Indian residential connection), team drive-throughs with timestamped photos and depth categories; IEC application; freeze the Kotlin app; fix Study 1 scoring (items 1–2); batch CLI (item 8).
Week 2–4	Depth-category reporting UI; traversal-silence logging with consent; per-edge harm term (item 3); Wilson/Beta bound (item 6); correlated-report discount (item 5); server-side LLM proxy (item 10); packed graph asset (item 9); Tamil Tier-0 templates.
Week 4–8	Pilot with one delivery fleet or dark-store cluster in the Velachery corridor; resident review of the watchlist; build a second, time-stamped corpus (2023 press and SAR, 2026 captures); re-run Studies 1–2 with event splits; Study 3 (300 routes $\times$ 3 tiers, human κ on a subsample); Study 4 on a Rs 10–15k phone.
Week 8–12	Release the Chennai benchmark (corpus + harness + baselines, Zenodo DOI); submit the workshop paper; fleet willingness-to-pay interviews (target: one manager naming a price above Rs 15,000/month); draft GCC/TNSDMA pilot note.

9 Research and publication plan

- **Paper 1 (now):** the attached IEEE draft — system design, bounded-deterrence result, fair replay evaluation, honest calibration. Suitable for a workshop or applications track (your notes name SIGSPATIAL ARIC, ISCRAM, IEEE ITSC).
- **Paper 2 (after the monsoon):** traversal silence as missing-not-at-random evidence, calibrated on a held-out event.
- **Paper 3:** verified explanation on low-cost phones — pass rate by tier and language, and appropriate reliance in a within-subjects study.
- **Benchmark:** the Chennai replay corpus and harness with a DOI; benchmarks are reused and cited more than systems.

10 The next ten actions

1. Turn on data capture scripts today (monsoon onset is imminent).
2. Patch Study 1 to score under a common reference belief; re-run at $N = 1,000$ with a batch CLI.
3. Add the per-edge harm term and re-run the sweep.
4. Add depth categories to the report screen.
5. Move the Groq key server-side.
6. Replace the JSON graph asset with a packed binary.
7. Buy or borrow the cheapest target phone and run Study 4.
8. Submit the IEC application.
9. Fill in the author names, affiliations and e-mails in the paper and literature review.
10. Book one fleet-operator conversation this week.

Appendix: verification

The literature review cites 129 works: 92 verified against Crossref records by DOI, 33 against arXiv abstract pages, and 4 against secondary listings (Beta reputation 2002, Goldberg–Harrelson SODA 2005, two patents). Twelve candidates were dropped (ten lookups rate-limited, two wrong DOIs). Numbers in this document come from your `data/results/*/result.json` files and from the re-analysis described in Section 4; the re-analysis code and outputs are included in the delivered bundle.